



Exam :

End Term Quiz

Subject :

Maths2

Total Marks :

50.00

QP :

2023 Apr30: IIT M DIPLOMA ET1 EXAM 2023 Apr30: IIT

Exam Mode

Learning Mode

View Question Paper Summary

QUESTION MENU

1	2	3	4	5	6	7	8	9
10	11	12	13	14	15	16	17	18
19	20	21	22	23	24	25	26	27
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TIMER

00:20



CONTROLS

SUBMIT EXAM

Your Score

0.00 / 50.00

(0%)

Question 1 : 640653565959

Total Mark : 0.00 | Type : MCQ

THIS IS QUESTION PAPER FOR THE SUBJECT "**MATHEMATICS FOR DATA SCIENCE 2 (COMPUTER BASED EXAM)**" ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT? CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE

WRITTEN. (IF IT IS NOT THE CORRECT SUBJECT ,PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

OPTIONS :

☐ YES

☐ NO

Your score : 0

 Discussions (0)



Question 2 : 640653565960

 View Solutions (0)

Total Mark : 2.00 | Type : SA

If $A = \begin{pmatrix} a+b+c & b+c-d \\ b-c+1 & d \end{pmatrix} = \begin{pmatrix} a+1 & 2c-3d+3 \\ 2 & c+a \end{pmatrix}$, then what is the value of $tr(A)$, where $tr(A)$ denotes the trace of A (sum of diagonal entries of A)?



Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

 Discussions (0)



Question 3 : 640653565961

Total Mark : 0.00 | Type : COMPREHENSION

Let D denote the set of all 2×2 diagonal matrices. 

Consider an ordered basis $\beta = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix} \right\}$.

Let $T : U \rightarrow \mathbb{R}^2$ be a linear transformation defined as

$T \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} = (a + b, 4a - 5b)$. Let matrix A be the

matrix representation of T with respect to the

ordered basis β for D and the standard ordered

basis for the co-domain $\mathbb{R}^2$.

Answer the given subquestions.

Your score : 0



Question 4 :

640653565962

 View Parent QN

 View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Which of the following matrix is A?

OPTIONS :

☐ $\begin{pmatrix} 1 & 4 \\ 2 & -10 \end{pmatrix}$ 

☐ $\begin{pmatrix} 1 & 4 \\ 1 & -5 \end{pmatrix}$ 

☐ $\begin{pmatrix} 1 & 2 \\ 4 & -10 \end{pmatrix}$ 

☐ $\begin{pmatrix} 1 & 1 \\ 4 & -5 \end{pmatrix}$ 

Your score : 0

[Discussions \(0\)](#)**Question 5 :****640653565963**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : MSQ

Which of the following is/are true about T ?

OPTIONS :

☐ Rank of T is 2. ☐ There exists an isomorphism from $R(T)$ (range of T) to $\mathbb{R}$. ☐ Nullity of T is 1. ☐ T is one-one.

Your score : 0

[Discussions \(0\)](#)**Question 6 :****640653565964**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : MSQ

Let B be the matrix representation of T with respect to the ordered basis β for the domain and the ordered basis $\{(1, -1), (1, 1)\}$ for the co-domain $\mathbb{R}^2$. Which of the following is/are true?



OPTIONS :

- ☐ A is similar to B . 
- ☐ Rank of B is 1. 
- ☐ Nullity of B is 0. 
- ☐ A is not similar to B . 

Your score : 0

 Discussions (0)



Question 7 : 640653565965

Total Mark : 0.00 | Type : COMPREHENSION

Consider the subspace $W = \{(x, y, z) \mid x - y + z = 0, x, y, z \in \mathbb{R}\}$ of $\mathbb{R}^3$. Answer the given subquestions based on the given information. 

Your score : 0



Question 8 : 640653565966

 View Parent QN

 View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Which of the following is a basis β for W ?

OPTIONS :

- ☐ $\beta = \{(1, 1, 0), (-1, 0, 1)\}$ 
- ☐ $\beta = \{(1, 1, 0), (0, 1, -1)\}$ 
- ☐ $\beta = \{(0, 1, 1), (1, 1, 0), (-1, 0, 1)\}$ 
- ☐ $\beta = \{(0, 1, 1), (0, 1, -1)\}$ 

Your score : 0

Discussions (0)



Question 9 :

640653565967



View Parent QN



View Solutions (0)

Total Mark : 4.00 | Type : SA

If γ is the orthonormal basis of W obtained from β (from the previous question) by using the Gram Schmidt process with respect to the usual inner product, and (a, b, c) is the projection of $(1, 0, 1)$ onto W , then what is $6(a + b + c)$?



Answer (Numeric):

Answer

Accepted Answer : 8

Your score : 0

Discussions (0)



Question 10 : 640653565972

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Let $f(x, y) = x^4 + y^4 + 4x^3 + 4x^2 - 8y^2 - 5$.



Your score : 0



Question 11 :
640653565973

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 3.00 | Type : SA

The number of critical points of $f(x, y)$ is

Answer (Numeric):

Accepted Answer : 9

Your score : 0

[Discussions \(0\)](#)

Question 12 :
640653565974

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

The number of saddle points of $f(x, y)$ is

Answer (Numeric):

Accepted Answer : 4

Your score : 0

[Discussions \(0\)](#)

Question 13 :
640653565975

[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

The number of local maxima of $f(x, y)$ is

Answer (Numeric):

Answer

Accepted Answer : 1

Your score : 0

 Discussions (0)



Question 14 :
640653565976



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

The number of local minima of $f(x, y)$ is

Answer (Numeric):

Answer

Accepted Answer : 4

Your score : 0

 Discussions (0)



Question 15 : 640653565978

Total Mark : 0.00 | Type : COMPREHENSION

From the list of given terms find out the best possible options for each of the given subquestions:

- 1) Rank 2
- 2) Zero nullity
- 3) Non-zero determinant
- 4) Closure with respect to addition and scalar multiplication
- 5) Existence of zero element
- 6) Existence of additive inverse
- 7) Commutativity of addition
- 8) Elements
- 9) Associativity of addition
- 10) Global maxima
- 11) Saddle point
- 12) Global Minima
- 13) Gradient exists
- 14) Directional derivative exists in any direction
- 15) Orthonormal columns
- 16) First order partial derivatives exist
- 17) Function is differentiable
- 18) Affine subspace.
- 19) Limit exists



Your score : 0



Question 16 :

640653565979

View Parent QN

View Solutions (0)

Total Mark : 2.00 | Type : SA

A critical point can be _____. (Enter 3 best possible options. Enter only the serial numbers of those options in increasing order without adding any comma or space in between them.) [Suppose your answer is 7, 14 and 17, then you should enter 71417]

Answer (Numeric):

Answer

Accepted Answer : 101112

Your score : 0

[Discussions \(0\)](#)**Question 17 :****640653565980**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

An invertible matrix of order 3 has _____. (Enter 2 best possible options. Enter only the serial numbers of those options in increasing order without adding any comma or space in between them.) [Suppose your answer is 7 and 17, then you should enter 717]

Answer (Numeric):

Answer

Accepted Answer : 23**Your score : 0**[Discussions \(0\)](#)**Question 18 :****640653565981**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 1.00 | Type : SA

If the tangent plane exists for a function at a point then at that point _____. (Enter 5 best possible options. Enter only the serial numbers of those options in increasing order without adding any comma or space in between them.) [Suppose your answer is 1, 2, 3,4 and 5, then you should enter 12345]

Answer (Numeric):

Answer

Accepted Answer : 1314161719**Your score : 0**[Discussions \(0\)](#)

Question 19 :
640653565982



View Parent QN



View Solutions (0)

Total Mark : 1.00 | Type : SA

Some of the conditions that need to be checked to identify a vector space V are: _____ (Enter 5 best possible options. Enter only the serial numbers of those options in increasing order without adding any comma or space in between them.) [Suppose your answer is 7, 14, 11, 15 and 17, then you should enter 714111517]

Answer (Numeric):

Answer

Accepted Answer : 45679

Your score : 0



Discussions (0)



Question 20 : 640653565968



View Solutions (0)

Total Mark : 2.00 | Type : SA

What is the minimum sum of three non-negative numbers whose product is 27?

Answer (Numeric):

Answer

Accepted Answer : 9

Your score : 0



Discussions (0)



Question 21 : 640653565971



View Solutions (0)

Total Mark : 2.00 | Type : SA

Let V_1, V_2 and V_3 represent the subspaces $\{(x, y, z) : 2x + 3y + 4z = 0\}$, $\{(x, y, z) : x + 2y + z = 0\}$ and $\{(x, y, z) : x + y + 2z = 0\}$, respectively of $\mathbb{R}^3$. Let $A_i, i = 1, 2, 3$ be the affine spaces corresponding to $V_i, i = 1, 2, 3$. Suppose A_1 contains $(1, 1, -2)$, A_2 contains $(-2, 1, 1)$ and A_3 contains $(1, -2, 1)$. If (x_1, x_2, x_3) is the point of intersection of A_1, A_2 and A_3 , find $x_1 + 2x_2 + x_3$.



Answer (Numeric):

Accepted Answer : 1

Your score : 0

 Discussions (0)

Question 22 : 640653565969

 View Solutions (0)

Total Mark : 1.00 | Type : SA

Find the number of values of a so that the system of linear equations 

$$x + y - z = 0, 2x + (a + 2)y + z = 0, ax + y + z = 0$$
 has infinitely many solutions.

Answer (Numeric):

Accepted Answer : 2

Your score : 0

 Discussions (0)

Question 23 : 640653565970

 View Solutions (0)

Total Mark : 1.00 | Type : MSQ

Let A be a 3×3 orthogonal matrix with positive determinant. Which of the following option(s) is/are true?

OPTIONS :

- ☐ The determinant of $2A$ is 8.
- ☐ The determinant of $3A$ is 3.
- ☐ A is equivalent to the identity matrix of order 3.
- ☐ None of these

Your score : 0

Discussions (0)



Question 24 : 640653565977

View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Consider the function $f(x, y) = x^2 + xy$. Which of the following line represents the tangent line at the point $(1, 1)$ in the direction of the vector $(1, 1)$?

OPTIONS :

☐ $\{(x, y, z) \in \mathbb{R}^3 \mid \frac{x-1}{\frac{1}{\sqrt{2}}} = \frac{y-1}{\frac{1}{\sqrt{2}}} = \frac{z-2}{2\sqrt{2}}\}$

☐ $\{(x, y, z) \in \mathbb{R}^3 \mid \frac{x-1}{\frac{1}{\sqrt{2}}} = \frac{y-1}{\frac{1}{\sqrt{2}}} = \frac{z-1}{\frac{1}{\sqrt{2}}}\}$

☐ $\{(x, y, z) \in \mathbb{R}^3 \mid \frac{x-1}{\frac{1}{\sqrt{2}}} = \frac{y-1}{\frac{1}{\sqrt{2}}} = \frac{z-2}{\frac{3}{\sqrt{2}}}\}$

☐ $\{(x, y, z) \in \mathbb{R}^3 \mid \frac{x}{\frac{1}{\sqrt{2}}} = \frac{y}{\frac{1}{\sqrt{2}}} = \frac{z}{2\sqrt{2}}\}$

Your score : 0

Discussions (0)



Question 25 : 640653565983

View Solutions (0)

Total Mark : 3.00 | Type : MSQ

Choose the correct statements for the function

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^4 + y^4} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

OPTIONS :

☐ $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \frac{4}{17}$ along the path $y = 2x$

☐ $f_x(0,0) = 1$

☐ The directional derivative of f at $(0,0)$ in the direction of $(1,-1)$ exists.

☐ f is not continuous at $(0,0)$.

Your score : 0

Discussions (0)



Question 26 : 640653565984

[View Solutions \(0\)](#)

Total Mark : 4.00 | Type : MCQ

Match the equation of the surface in Column A with the tangent plane at the point $(1, 1, 2)$ in column B and the vector subspace corresponding to the affine subspace (of $\mathbb{R}^3$) formed by the tangent plane, in Column C.

	Equation of the surface (Column A)		Equation of the tangent plane at $(1, 1, 2)$ (Column B)		Vector subspace corresponding to the affine subspace formed by tangent plane (Column C)
i)	$z = x^2y + xy^2$	a)	$3x + 3y - z = 4$	1)	$\{(x, y, z) \mid 6x = 2y + z, x, y, z \in \mathbb{R}\}$
ii)	$z = 3x^2 - y^2$	b)	$z = -2x - 2y + 6$	2)	$\{(x, y, z) \mid x + y = \frac{1}{3}z, x, y, z \in \mathbb{R}\}$
iii)	$xyz = 2$	c)	$6x - 2y = z + 2$	3)	$\{(x, y, z) \mid z = -2(x + y), x, y, z \in \mathbb{R}\}$

Table: M2ES1

Choose the correct option from the following:

OPTIONS :

☐ i) $\rightarrow$ b $\rightarrow$ 1, ii) $\rightarrow$ a $\rightarrow$ 2, iii) $\rightarrow$ c $\rightarrow$ 3

☐ i) $\rightarrow$ a $\rightarrow$ 2, ii) $\rightarrow$ c $\rightarrow$ 1, iii) $\rightarrow$ b $\rightarrow$ 3

- ☐ i) $\rightarrow b \rightarrow 1$, ii) $\rightarrow c \rightarrow 3$, iii) $\rightarrow a \rightarrow 2$ 
- ☐ i) $\rightarrow b \rightarrow 1$, ii) $\rightarrow c \rightarrow 2$, iii) $\rightarrow a \rightarrow 3$ 

Your score : 0

 Discussions (0)



Question 27 : 640653565985

Total Mark : 0.00 | Type : COMPREHENSION

Consider the system of linear equations formed by the equations in column B of Table M2ES2 and let A be its coefficient matrix. Answer the given related subquestions.

	Equation of the surface (Column A)		Equation of the tangent plane at (1, 1, 2) (Column B)		Vector subspace corresponding to the affine subspace formed by tangent plane (Column C)
i)	$z = x^2y + xy^2$	a)	$3x + 3y - z = 4$	1)	$\{(x, y, z) \mid 6x = 2y + z, x, y, z \in \mathbb{R}\}$
ii)	$z = 3x^2 - y^2$	b)	$z = -2x - 2y + 6$	2)	$\{(x, y, z) \mid x + y = \frac{1}{3}z, x, y, z \in \mathbb{R}\}$
iii)	$xyz = 2$	c)	$6x - 2y = z + 2$	3)	$\{(x, y, z) \mid z = -2(x + y), x, y, z \in \mathbb{R}\}$

Table: M2ES2

Your score : 0



Question 28 : 640653565986

 View Parent QN

 View Solutions (0)

Total Mark : 3.00 | Type : MCQ

Which of the following denotes the solution space of this system of linear equations?

OPTIONS :

- ☐ $\{(x, y, 2) \mid x + y = 2, x, y \in \mathbb{R}\}$ 
- ☐ $\{(x, y, 2) \mid x, y \in \mathbb{R}\}$ 
- ☐ $\{(x, x, 2) \mid x \in \mathbb{R}\}$ 
- ☐ $\{(1, 1, 2)\}$ 

Your score : 0

 Discussions (0)



Question 29 :
640653565987

 View Parent QN

 View Solutions (0)

Total Mark : 2.00 | Type : MCQ

Which of the following sets denotes nullspace(A)?

OPTIONS :

- ☐ $\{(x, y, 0) \mid x, y \in \mathbb{R}\}$ 
- ☐ $\{(x, -x, 0) \mid x \in \mathbb{R}\}$ 
- ☐ $\{(x, x, 0) \mid x \in \mathbb{R}\}$ 
- ☐ $\{(0, 0, 0)\}$ 

Your score : 0

 Discussions (0)



Question 30 : 640653565988

Total Mark : 0.00 | Type : COMPREHENSION

Consider the system of linear equations formed by the equations in column B of Table M2ES3 and let A be its coefficient matrix. Answer the given related questions.

	Equation of the surface (Column A)		Equation of the tangent plane at (1, 1, 2) (Column B)		Vector subspace corresponding to the affine subspace formed by tangent plane (Column C)
i)	$z = x^2y + xy^2$	a)	$3x + 3y - z = 4$	1)	$\{(x, y, z) \mid 6x = 2y + z, x, y, z \in \mathbb{R}\}$
ii)	$z = 3x^2 - y^2$	b)	$z = -2x - 2y + 6$	2)	$\{(x, y, z) \mid x + y = \frac{1}{3}z, x, y, z \in \mathbb{R}\}$
iii)	$xyz = 2$	c)	$6x - 2y = z + 2$	3)	$\{(x, y, z) \mid z = -2(x + y), x, y, z \in \mathbb{R}\}$

Table: M2ES3

Your score : 0

**Question 31 :****640653565989**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : SA

Find Rank(A).

Answer (Numeric):

Answer

Accepted Answer : 3

Your score : 0

[Discussions \(0\)](#)**Question 32 :****640653565990**[View Parent QN](#)[View Solutions \(0\)](#)

Total Mark : 2.00 | Type : MSQ

Which of the following denotes the column space of A?(More than one option may be correct)

OPTIONS :

☐ $\mathbb{R}^3$ 

☐ $\text{Span}\{(-1, -1, 1)\}$ 

☐ $\text{Span}\{(3, 6, 2), (3, -2, 2)\}$ 

☐ $\text{Span}\{(3, 6, 2), (3, -2, 2), (-1, -1, 1)\}$ 

Your score : 0

 Discussions (0) SUBMIT EXAM